

* Skewness
easy

1) 50, 55, 60, 65, 70

$$\text{Mean} = 60$$

$$\text{Median} = 60$$

Skewness = No skewness (symmetrical)

2) 1, 2, 3, 4, 20

$$\text{Mean} = 6$$

$$\text{Median} = 3$$

Skewness = positive skewness

Median

1) 100, 102, 104, 106, 108, 110, 70

70, 100, 102, 104, 106, 108, 110

$$\text{Mean} = 100$$

$$\text{Median} = 104$$

Skewness = Negative skewness

2) 8, 9, 10, 11, 12, 13, 40

$$\text{Mean} = 14.71$$

$$\text{Median} = 11$$

Skewness = Positive

Hard

31

1) 10, 15, 20, 25, 30, 35, 40, 100

Mean = 39.29

Median = 27.5

Skewness = Positive

2) 80, 82, 84, 86, 88, 90, 92, 20

20, 80, 82, 84, 86, 88, 90, 92

Mean = 77.75

Median = 85

Skewness = Negative

* Frequency level

1) 5, 10, 5, 15, 20, 10, 5, 20, 25, 15

x value	f (frequency)	cumulative of
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5	3	3	15
10	2	5	20
15	2	7	30
20	2	9	40
25	1	10	25

		1	26
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5, 5, 5, 10, 10, 15, 15, 20, 20, 25

Mean : 12.5
 Median : 12.5
 skewness : positive

2) 12, 18, 15, 22, 25, 28, 31, 35, 38, 42,
 45, 48, 52, 55, 58, 62, 65, 68, 72, 75,
 78, 82, 85, 88, 92, 95, 14, 19, 24, 27,
 33, 37, 41, 46, 49, 53, 57, 61, 66, 69,
 73, 77, 81, 84, 89, 91, 96, 98, 100, 17

x f Σf cumulative $\Sigma x.f$

10-19	6	6	87
20-29	5	11	122.5
30-39	5	16	172.5
40-49	6	22	267
50-59	5	27	272.5
60-69	6	33	387
70-79	5	38	372.5
80-89	6	44	507
90-100	6	50	570

2758.0

$$\text{Mean} = \frac{2758.0}{50} = 55.16$$

$$\text{median} = L + \left(\frac{\frac{n}{2} - cf}{f} \right) \times h$$

$$\text{Lower} = 50 - 0.5 = 49.5$$

$$\text{upper} = 50 + 0.5 = 50.5$$